Reading Activity 3: Feature Selection for Regression Problems

CCELEC2L
COM232
Lorenzo Miguel I. Vallefas

Introduction

Student performance depends on a mix of academic, social, and personal factors. Not all variables are equally informative, and including too many can add noise, increase model complexity, and make results harder to interpret.

This analysis aims to predict secondary-level students' final mathematics grade (G3) while identifying the features that matter most. Two methods are used to select important variables and focus the model on meaningful predictors, making it simpler and easier to understand which factors affect student performance.

Dataset Overview

The dataset contains a wide range of features describing student background, family environment, academic behavior, and lifestyle factors. These variables are meant to capture the different conditions under which secondary-level students learn and perform in mathematics.

The table below summarizes the input features used in this study along with their descriptions. The (**G3**) is treated as the target variable:

Feature Name	Type	Description
school	Categorical	School attended by the student (Gabriel Pereira or Mousinho da Silveira)
sex	Binary	Student’s gender
age	Integer	Student’s age (15–22 years)
address	Binary	Home address type (urban or rural)
famsize	Binary	Family size (≤3 members or >3 members)
Pstatus	Binary	Parents’ cohabitation status (together or apart)
Medu	Integer	Mother’s education level (0–4 scale)
Fedu	Integer	Father’s education level (0–4 scale)
Mjob	Categorical	Mother’s occupation
Fjob	Categorical	Father’s occupation
reason	Categorical	Reason for choosing the school
guardian	Categorical	Student’s primary guardian
traveltime	Integer	Travel time from home to school (1–4 scale)
studytime	Integer	Weekly study time (1–4 scale)
failures	Integer	Number of past class failures
schoolsup	Binary	Extra educational support from school
famsup	Binary	Educational support from family
paid	Binary	Participation in paid extra classes
activities	Binary	Participation in extracurricular activities
nursery	Binary	Attendance in nursery school
higher	Binary	Intention to pursue higher education
internet	Binary	Internet access at home
romantic	Binary	Involvement in a romantic relationship
famrel	Integer	Quality of family relationships (1–5 scale)
freetime	Integer	Amount of free time after school (1–5 scale)
goout	Integer	Frequency of going out with friends (1–5 scale)
Dalc	Integer	Workday alcohol consumption (1–5 scale)
Walc	Integer	Weekend alcohol consumption (1–5 scale)
health	Integer	Current health status (1–5 scale)
absences	Integer	Number of school absences
G1	Integer	First period mathematics grade
G2	Integer	Second period mathematics grade

Dataset Analysis

To better understand the dataset and choose the most relevant features, Exploratory data analysis techniques will be conducted.

A. Final Grade Distribution

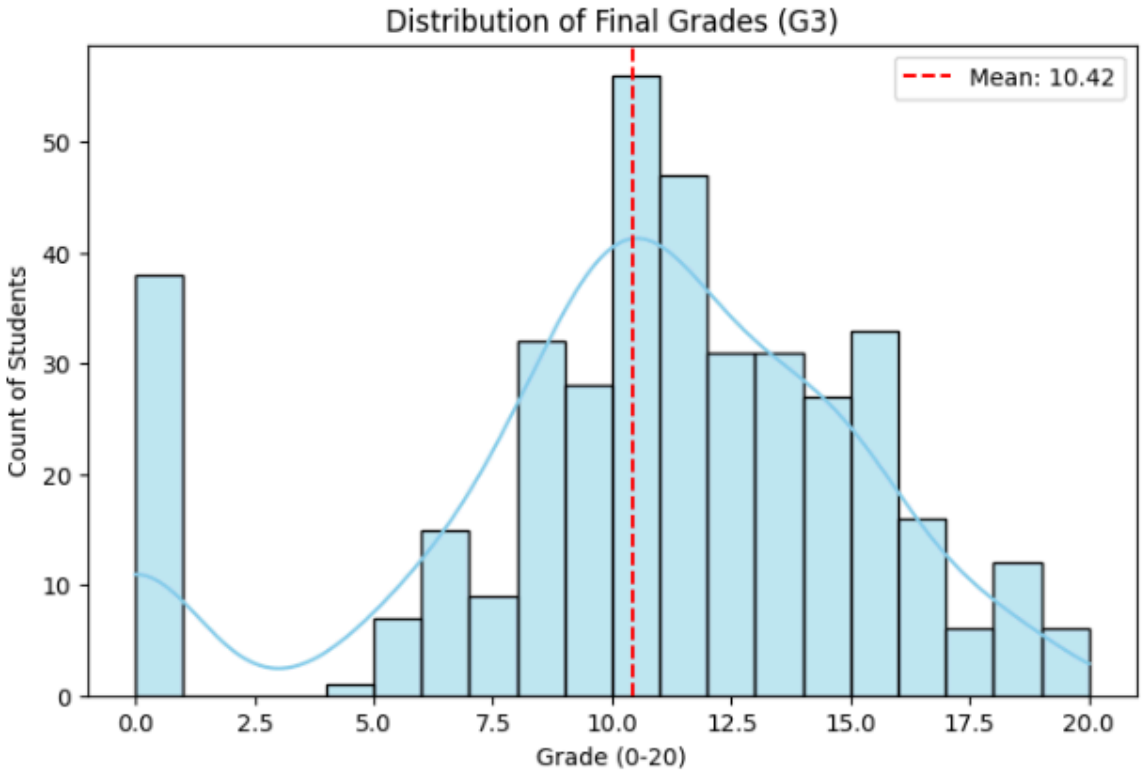

Figure 1

In **Figure 1**, Most students have final grades between 8 and 14, indicating an average performance range, but the dataset has a relatively small sample size, so conclusions should be interpreted carefully. A few students scored 0, which likely reflects absences or dropouts.

B. Correlation Matrix

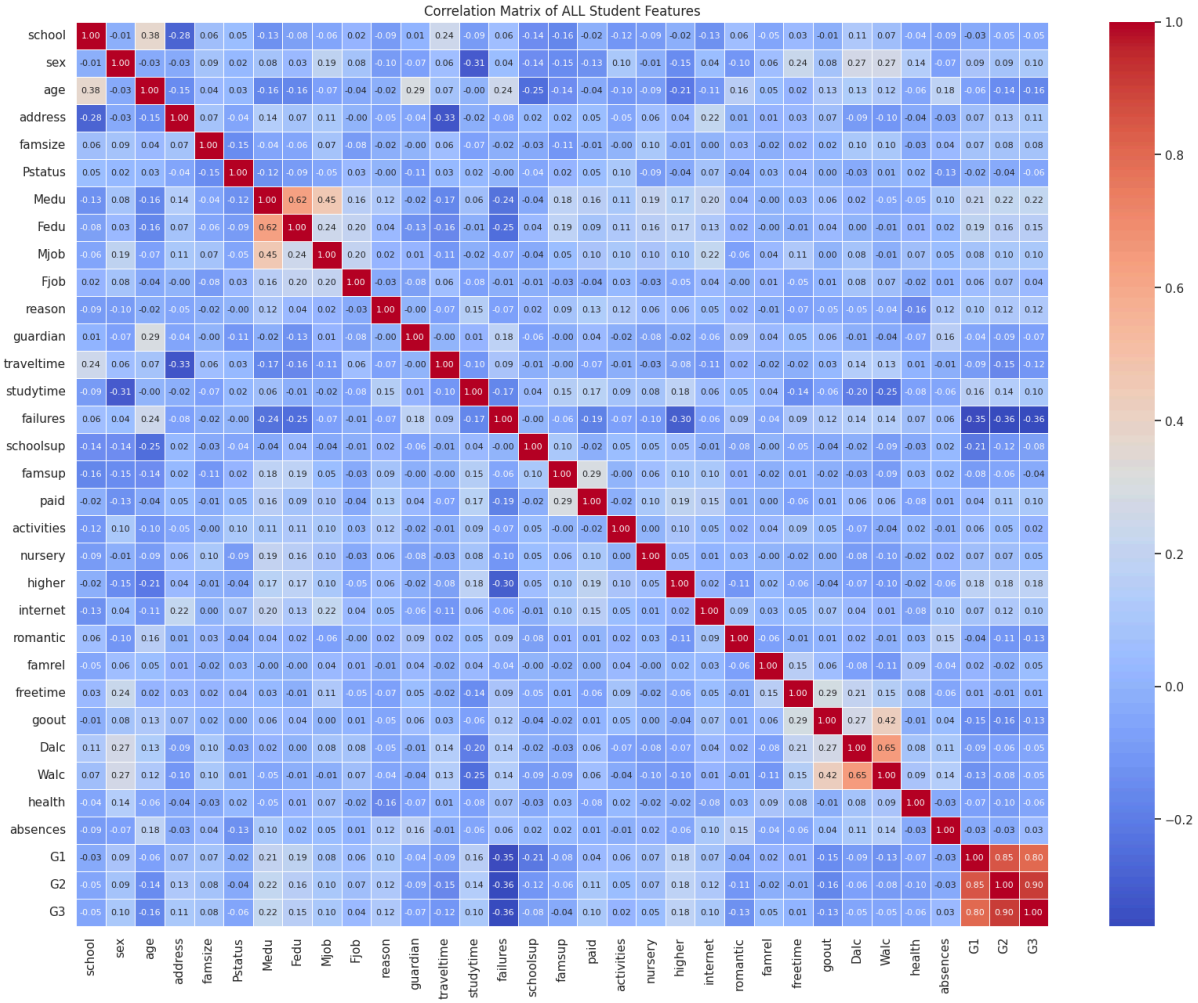

Figure 2

Figure 2 shows how different factors relate to final grades (G3). Previous grades, G2 (0.905) and G1 (0.801), are the strongest predictors, showing that performance tends to be consistent over time. Parental education (Medu 0.217, Fedu 0.152) and students' plans for higher education (higher 0.182) also help improve grades. Minor positive influences include reason for school choice, address, sex, mother's job, and paid classes. On the negative side, past failures (-0.360) are the biggest warning sign, while older age (-0.162), frequent socializing (-0.133), being in a relationship (-0.130), and long commutes (-0.117) slightly lower performance. Weaker negative factors like school support, guardian, health, family status, and weekday alcohol consumption have minimal impact. In short, grades are mainly defined by academic history, family background, motivation, and some lifestyle choices.

Problem Statement

While the dataset contains many variables describing students' academic, social, and environmental characteristics, including all of them may only introduce unnecessary information, potentially confusing the model and reducing the accuracy of its predictions since by only successfully selecting informative features can significantly increase learning accuracy and improve result [1]. Furthermore, Machine learning models that include all available features without proper selection are prone to overfitting, increased prediction errors, bias, and poor generalization. By focusing on a carefully selected set of features, models can generalize better to new data and deliver more reliable performance in real-world scenarios [2].

Feature Selection

To ensure the predictive model focuses on the most meaningful variables, this study compares two feature selection strategies.

A. Literature-base Approach

Student performance is affected by several measurable factors. Past class failures show a strong negative impact, meaning students with previous failures tend to achieve lower final grades. Age also has a slight negative effect, with older students performing worse on average. Positive influences include maternal education and the student's aspiration to pursue higher education, both of which are associated with higher grades [3].

Past Class Failures	Student Age	Mother's Education	Intention for higher education

Table 1: Factors affecting students grade base on past literature

B. Data-driven Approach

The data-driven statistical feature set focuses on what the dataset itself indicates as important for predicting student grades (G3). Numerical features were selected based on their correlation with G3, while categorical features were converted into numerical features as well, comparing the average G3 across categories to identify meaningful differences. Features with little variation or minimal contribution were excluded.

Feature	Type	Score / Value
failures	Numerical	-0.3604
Medu	Numerical	0.2171
age	Numerical	-0.1616
Fedu	Numerical	0.1524
higher	Categorical	0.152457
Goout	Numerical	-0.1327

Table 2: Selected features affecting student grades

The features G1 and G2 were excluded in the feature selection despite showing high correlation because relying heavily on past grades weakens the value of model training. It essentially results in predicting final grades based on prior grades rather than on independent influencing factors.

Model Training

After selecting the features, both approaches will be trained using a Random Forest Regressor. Once training is complete, their performance will be evaluated using Mean Absolute Error (MAE), which measures the average difference between predicted and actual grades, and R^2 (R-squared), which indicates how well the model explains the variability of the target variable [4].

A. **R2**

Figure 3

Figure 3 shows Set Literature-based outperformed Data Driven. The negative R2 in Data Driven shows that including Fedu and goout added noise and redundancy. Removing these weak predictors in Literature-based improved the R2, showing that focusing on factors like academic history (failures) and motivation (higher) helps the Random Forest capture patterns more effectively.

B. **Mean Absolute Error**

Figure 4

With grades ranging from 0 to 20, an MAE of 3.6 means predictions are off by about 3–4 points. Literature-base slightly outperforms Data Driven, indicating that removing less relevant features improved efficiency. However, prediction accuracy remains limited due to other potentially important features not being included and the relatively small size of the dataset. Overall, Literature-based is the more reliable and stable predictor.

Conclusion

Simplicity could possibly beat complexity: Literature-base, with fewer but more relevant features, outperformed the more complex Data Driven.

Demographics alone are limited: Using only demographic data provides poor predictions. Accurate forecasting of student performance requires current academic information. A student's background alone is not enough.

Personal Insight

The dataset is too small and imbalanced to fully capture patterns in student grades. Exploring additional features could improve predictions, but accurately forecasting grades is inherently difficult, since each student's performance depends on a unique combination of factors which may still be unexplored in this dataset.

REFERENCES

[1]X. Li, Y. Wang, and R. Ruiz, "A Survey on Sparse Learning Models for Feature Selection," *IEEE Transactions on Cybernetics*, vol. 52, no. 3, pp. 1642–1660, May 2020, doi: https://doi.org/10.1109/tcyb.2020.2982445.

[2]M. Uddin, J. Lee, S. Rizvi, and S. Hamada, "Proposing Enhanced Feature Engineering and a Selection Model for Machine Learning Processes," *Applied Sciences*, vol. 8, no. 4, p. 646, Apr. 2018, doi: https://doi.org/10.3390/app8040646.

[3]N. R. Beckham, L. J. Akeh, G. N. P. Mitaart, and J. V. Moniaga, "Determining factors that affect student performance using various machine learning methods," *Procedia Computer Science*, vol. 216, pp. 597–603, 2023, doi: https://doi.org/10.1016/j.procs.2022.12.174.

[4]GeeksforGeeks, "Regression Metrics," *GeeksforGeeks*, Oct. 12, 2023. https://www.geeksforgeeks.org/machine-learning/regression-metrics/